

Booth encode

Radix-four Booth recodes the multiplier Y into signed digits zero, plus or minus one, or plus or minus two

0C times 1A starter

- Starter: X equals hex zero C, twelve signed; Y equals hex one A, twenty-six
- Encode Y into four Booth digits: minus two, minus one, plus two, zero
- Digit zero triplet one-zero-zero uses y minus one equals zero
- Partial products: minus twenty-four, minus forty-eight, plus three eighty-four, zero
- Booth sum and signed X times Y both equal three twelve
- Click Encode Y to see the bit grid, digit cards, and partial-product list

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Booth encode grid

Radix-4 Booth: overlapping triplets of the multiplier encode to $\{0, \pm 1, \pm 2\}$ — fewer partial products than bit-by-bit.

Starter example: $X=0x0C$ (12), $Y=0x1A$ (26) — encode Y into radix-4 Booth digits and form partial products.

Load starter example

Challenges (1/22)

Quiz: Booth. Multiplier recoding family name? Answer: Booth

Answer

Show hint

Check

Next

Idle

quiz-booth

quiz-radix4

quiz-digits

quiz-half

starter

encode

example

neg-digit

plus2

zero-y

product-match

demo

explain

swap

triplet0

code-table

digit-count

shift-step

booth-000

edit-y

neg-pp

full-path

Core ideas

Triplets

Overlap by one bit; include $y[-1]=0$ for the first group.

Digits

Each triplet $\rightarrow 0, \pm 1$, or ± 2 (not ± 3).

Fewer PPs

$n/2$ partial products vs n for radix-2.

Booth encode starter

learn_digital — booth encode

Workbook practice

- On paper, encode Y equals one A into four radix-four Booth digits using the cheat table
- For triplet zero-one-one, record digit plus two
- Compute partial products for X equals twelve and sum them
- Draw how triplets overlap, each group shares one bit with the next
- Name one pitfall: forgetting y minus one on the first digit

Pitfalls to watch

- Do not treat Booth digits as plain binary, they are signed recoding
- Triplet bits are y_{2i+1} , y_{2i} , y_{2i-1} , not independent pairs
- Partial products with minus one or minus two need signed X
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, encode the starter and confirm product three twelve
- On paper, fill one row of the Booth table from a triplet you choose
- When you are ready, take the short quiz, then continue to signed arithmetic